

TARIK BULUT

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Education

Charles Darwin University

Master Of Information Technology Artificial Intelligence GPA - 6/7

Nov. 2024 – Nov 2026

Darwin, Australia

Firat University

Faculty of Technology Software Engineering GPA - 3.26/4

Aug. 2019 – Jul 2023

Elazığ, Turkey

Experience

North Australia Centre for Autonomous Systems (NACAS)

Casual Professional

Mar 2026 – Present

Darwin, NT, Australia

- Developing end-to-end **computer vision pipelines** and **AI models** for industrial applications, encompassing **data collection, preprocessing, and model deployment**.
- Collaborating with a research team to design and optimize **autonomous systems** for real-world environmental and **industrial monitoring** tasks.
- Leveraging **Python** and **deep learning frameworks** to enhance the accuracy and efficiency of **automated data analysis** and monitoring systems.

Procusale

Remote Intern / Volunteer

Nov 2021 – Nov 2023

Izmir, Turkey

- Developed various **web and mobile applications** using **Flutter, Firebase, and WordPress** for diverse clients including NGOs and law firms.
- Designed **UI/UX prototypes** in **Figma** and managed cloud deployments on **DigitalOcean**.

Projects

IT Code Fair Projects – Dark Sky NT & Kriol Translation | Python, PyTorch, Hugging Face

Nov 2025

- Dark Sky NT:** Developed a stargazing recommendation platform using multi-source data pipelines; implemented a fine-tuned **multilingual BERT** model for sentiment analysis and a **LoRA-adapted Qwen** model with **FAISS** for RAG-based recommendations.
- Kriol Translation:** Built an end-to-end **neural machine translation** system using **NLLB-200** to translate Kriol (Australian creole) into English, focusing on data augmentation for low-resource languages.
- Achievements:** Recognized as **Winner (AI Challenge)** and **Runner-Up (Data Science Challenge)** at CDU IT Code Fair 2025; Nominated for the **NT Digital Excellence Awards 2025**.

Connect4 & League of Legends Wiki MCP | PyTorch, PPO, MCP, FastAPI, Selenium

Jul 2025 – Aug 2025

- Leveraged **AI-aided coding (Claude)** to architect and manage complex system integrations, focusing on rapid prototyping and advanced modular design.
- Developed a **Connect4 RL agent** using **PPO** with self-play training and hybrid **CPU-GPU vectorized environments**.
- Built an **MCP server** providing AI agents with real-time access to **LoL Wiki data** (stats, runes, champions) to **mitigate hallucinations** and ensure accurate tactical recommendations.

Doom - Mario AI | Python, Stable Baselines, Gym

Jan 2023

- Developed AI agents capable of playing **Mario** and **Doom (1992)** using **PPO** and **DQN** algorithms.
- Conducted a comparative performance analysis of RL algorithms in diverse environments and **authored a technical paper** available on GitHub.

Technical Skills

Languages: Python, Java, C#, Dart, SQL, HTML/CSS

Libraries/Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, FastAPI, Hugging Face, Transformers

AI/ML Competencies: Computer Vision, Reinforcement Learning (PPO/DQN), Large Language Models, RAG (FAISS), Fine-tuning (LoRA), NLP (NLLB-200), Data Pipelines, LLM APIs, Prompt Engineering, LangChain, Vector Databases

Developer Tools & Cloud: Git, Docker, VS Code, DigitalOcean, Firebase, Android Studio, Figma, Selenium, BeautifulSoup, MCP Protocol, REST API

Honors, Languages & Interests

Honors: Nominee for **NT Digital Excellence Awards 2025**; Winner (AI) & Runner-Up (Data Science) at **CDU IT Code Fair 2025**

Languages: Turkish (Native), English (Professional), German (Elementary)

Interests: Complex Strategy Games (**Europa Universalis IV-V**), League of Legends